

## CARBON SEQUESTRATION BY MANGROVES OF GUJARAT, INDIA

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### ABSTRACT

Gujarat has the second largest mangrove cover (1058 sq km) of country. Mangrove being the major woody habitats forms the important carbon sinks in the coastal regions. Present work has examined the carbon sequestration by mangroves of Gujarat. Tree harvesting methods was used for estimation of total biomass (above and below ground) by stratified random sampling. Further, carbon sequestered in the soil (up to 30 cm depth) was also examined. A total of 8.116 million ton carbon has been sequestered by mangroves of Gujarat. Mangrove soils contribute more than mangrove plants in the overall carbon sequestration.

**KEYWORDS:** Gujarat, Mangrove, Carbon Sequestration, Climate Change, Soil Organic Carbon

### INTRODUCTION

Tropical forests in general are an important part of the global carbon cycle (Clark, 2001). Mangrove ecosystems are large and dynamic reservoir of carbon, which is an important part of global carbon cycle and a potential sink of atmospheric carbon dioxide (Matsui, et. al., 2010). Although the area covered by mangrove forests represent only a small fraction of the tropical forests, their position at the terrestrial-ocean interface and potential exchange with coastal ocean waters suggest that these forests make a unique contribution to total carbon cycle in coastal ocean (Twilley, 1992). The potential impact of mangroves on the coastal zone carbon dynamics has been a topic of debate in past decades (Steven, et. al. 2008). The contribution of coastal and marine ecosystems to mitigate climate change through carbon sequestration and storage indicate that these ecosystems can rival their terrestrial counterparts (Yee., 2010). Blue carbon sinks include the open oceans, kelp forests, salt marshes, sea grass beds, coral reefs and mangroves. Management of these blue carbon sinks are currently not accounted for in climate change policies and are excluded from national carbon inventories and international carbon payment schemes (Lasco, 2004).

The biomass assessment is very significant (Zhang, et. al., 2006) and it endeavors to have two fundamental sets of information i.e. resource availability and environment management. In forest biomass studies, two biomass units are used, fresh weight (Araujo, et. al., 1999) and dry weight (Aboal, et. al., 2005; Montagu, et. al., 2005). Lu, et. al., (2004) has mentioned three methods for biomass estimation viz. field measurement, remote sensing and GIS-based approach amongst them the field measurement has been considered to be accurate. Further, in remote sensing and GIS-based approach also field based data is required for validation. All the three methods i.e. harvest method (field measurement), mean tree method (remote sensing) and the allometric method (GIS method) have been used to estimate the biomass of different mangrove forests of the world. The mean tree method may be utilized only in forests with homogenous tree size and distribution such as plantation areas. In the case of allometric method, whole or partial weight of a tree is estimated with the help of measurable tree dimensions such as trunk diameter and height by using allometric equations. The allometric method is based on the assumption that one part of the organism is proportional to that of another. Therefore, the trunk diameter is

correlated with the trunk weight. However, these allometric equations have been reported to vary with the species and site as well (Cloug, et. al, 1997; Smith III & Whelan, 2006; Komiyana, et.al., 2008).

Most of the allometric equations of mangrove have been developed for single stemmed mangrove species. However, some allometric equations have also been developed for multi-stemmed mangrove species such as *Rhizophora*, *Avicennia* and *Excoecaria* (Cloug, et. al, 1997). Saenger (2002) has cited 43 allometric equations for the above ground biomass of mangroves. The species specific allometric equations (reported by different scientists) are significantly different for the same mangrove species (Tam, et. al, 1995; Saenger, 2002; Ong, et. al, 2004; Comley, et. al, 2005). Further, the question arises as to whether a species specific allometric equation may be applicable to different mangrove forests of the world. Ong, et.al., (2004) have found similar equation for two sites for *Rhizophora apiculata*. Clough, et.al., (1997) have reported different relationships at different mangrove sites. Therefore, allometric equations need to be developed for each site and species. Common allometric equations for mangroves have also been proposed by various scientists (Chave, et. al., 2005; Komiyama, et. al., 2005). These common allometric equations have been examined for various mangrove species (Komiyana, et. al., 2008) and were reported to show significant relative error with reference to the site specific and species specific equations.

Majority of the biomass estimation studies are focused on above-ground forest biomass (Brown, 1997; Kraenzel, et. al., 2003; Laclau, 2003; Losi, et.al., 2003; Aboal, et. al., 2005) because it accounts for majority of the total accumulated biomass in the forest ecosystem. Using this method, the amount of carbon sequestered in the biomass is generally taken as 50% of the dry biomass (Losi, et. al., 2003; Montagu, et. al., 2005). Komiyana et. al. (2008) have examined 23 publications of mangroves of last 50 years and found that only nine of them dealt with both above ground and below ground biomass.

Considering these facts, although destructive and very laborious, the present work has selected harvesting method for estimation of biomass (above and below ground) for understanding the carbon sequestered by mangroves of Gujarat.

## STUDY AREA

The state of Gujarat is located on the west coast of India. The average annual rainfall ranges from 2000mm (South Gujarat) to less than 400mm (North west-Kachchh). Its mangrove cover has been distributed over four regions i.e. Kachchh, Gulf of Kachchh, Saurashtra and South Gujarat (Figure 1.1) The state has the longest coastline of the country hosting the second largest (1048 sq km) mangrove cover (FSI, 2011). However, for the present work is not based on this estimation. Instead of this, the mangrove cover estimated by GEER has been taken for estimation of carbon sequestered in the mangrove forests of the state for the present work (Table 2). A total of 15 mangrove species have been reported from Gujarat (Table 1). However, the mangrove populations are largely represented by a single species i.e. *A. marina* and the distribution of mangrove species over four regions is not uniform (Pandey & Pandey 2009, Pandey & Pandey, 2010). Species diversity of mangroves is maximum in South Gujarat and minimum in Saurashtra (Table 1). It may be attributed to the hydrological and adaphic conditions.

## MATERIALS AND METHODS

For estimation of carbon sequestration by mangroves of Gujarat, the present work has considered two carbon pools i.e. vegetation biomass and soil. Further, the mangrove cover estimated by GEER - 912.47 sq km (unpublished) - has been taken (Table 2) for the assessment of mangrove biomass. The carbon sequestered in the litter has not been examined owing to the fact that the litter is generated by trees which is either decomposed or mineralized and mixed in soil transported by the tidal currents. Since, the carbon content of mangrove plants and soil have been estimated, the carbon

content of litter has not been examined. Stratified random sampling method was used for assessment of mangrove populations. Mangrove forests of all the four regions were classified into three density classes i.e. dense, moderate and sparse. Subsequently, a sample comprising 0.03% area of the state mangrove cover was taken for estimating the population values of each mangrove species. For this, 10X100 m plots were laid. A total of 316 such plots were laid for estimation of mangrove plants and recruits in the state (Table 3). Mangrove recruits and trees were categorized into five height classes (<1.5 m, 1.5 to 3m, 3 to 4 m, 4 to 5 m and > 5 m) and documented. The plants of less than 1.5m were treated as recruits.

For biomass estimation, nine out of the 15 mangrove species of the state were selected for uprooting (Table 1). Owing to the relatively rare occurrence of rest six mangrove species, they were not selected for harvesting (Table 1). Further, one representative of each height class (if available) was uprooted for the nine mangrove species and both the biomass (shoot and root) was reported. Shoot biomass comprised of stem, branches and leaves while root biomass represented by below ground and aerial roots. After uprooting each recruit/tree, it was segregated into trunk, branches, leaves, aerial roots and roots and the green weights of each part were taken immediately after uprooting. Dry weights of uprooted plants were taken for estimation of carbon content. For estimation of dry weight of uprooted tree, 500 gm sample of each part (trunk, branches, leaves, aerial roots and roots) were oven dried at 70°C for 48 hrs and respective dry weight were calculated. Further, these oven dried samples were powdered and analyzed using CHNS elemental analyzer (using combustion method at 600°C) at SICART laboratory, Vallabh Vidya Nagar, Anand Gujarat which gave the percent carbon content of each part (by dry weight). The carbon sequestration potential (amount of carbon sequestered per recruit/tree) calculated for various height classes of the nine mangrove species. These values were multiplied by the respective no. of recruit/trees of different height classes of nine mangrove species to give total carbon sequestration by mangrove plants. These values were converted to area level estimates by using the values of recruit and tree density of mangroves in various density classes of four mangrove regions.

For examining the carbon pool of mangrove soil, the soil samples were collected from 68 locations spread over Kachchh, Gulf of Kachchh and South Gujarat. Since climatic conditions of Saurashtra and Gulf of Kachchh are similar, the soil samples have been collected from only one region i.e. Gulf of Kachchh to represent both. For each location two set of samples corresponding to the depths 1-15 cm and 16-30 cm have been collected. Therefore, a total of 136 soil samples, have been collected (Table 3). These soil samples were analyzed for soil density, bulk density, carbon content as per the standard method (Pansu, 2006) at laboratory of GEER Foundation.

The tree level estimates of carbon sequestration by mangrove trees were converted into area level estimates by using the data about mangrove tree and recruit density in different types of mangrove habitats. This has been aggregated with the area level estimation of soil carbon to give an estimate of the total carbon sequestered by the mangrove habitat.

## RESULTS

A total of 13644 lakhs recruits were estimated for mangroves of Gujarat of which 6658 lakh (49%), 678 lakhs (5%), 297 lakh (2%) and 6009 (44%) are from Kachchh, Gulf of Kachchh, Saurashtra and South Gujarat respectively (Table 4). For Kachchh, the recruit density is 16705 pa ha, 11788 per ha and 4139 per ha for the dense, moderate and sparse mangroves respectively. It was 5727 per ha, 2522 per ha and 914 per ha for the dense, moderate and sparse mangroves of Gulf of Kachchh respectively. In the case of Saurashtra, it was 10555 per ha, 7803 per ha and 2143 per ha for dense, moderate and sparse mangroves respectively. Interestingly the recruit density of the dense and sparse mangroves of South Gujarat was 58845 per ha and 16567 per ha respectively. Moderate type of mangrove forest is not found in South Gujarat region. Considering all the three density classes together, the recruit density comes out to be 11890 per ha, 3839

per ha, 8625 per ha and 41482 per ha for Kachchh, Gulf of Kachchh, Saurashtra and South Gujarat. Considering all the four regions and the three density classes together, it is 14905 for mangroves of Gujarat.

A total of 5377 lakh mangrove trees/bushes were estimated for Gujarat of which 4612 lakh, 566 lakh, 94 lakh and 104 lakh are in Kachchh, Gulf of Kachchh, Saurashtra and South Gujarat respectively. In terms of density classes, 3687 lakhs, 1009 lakhs and 654 lakhs are in the dense, moderate and sparse mangrove areas respectively (Table 5). The tree density of Kachchh is 11737 per ha, 6578 per ha and 4036 per ha for the dense, moderate and sparse mangroves respectively. In the case of Gulf of Kachchh, it is 5383 per ha, 1304 per ha and 216 per ha for the dense, moderate and sparse forests respectively. For the mangroves of Saurashtra, it was 4216 per ha, 576 per ha and 238 per ha for the dense, moderate and sparse mangroves respectively. In the case of South Gujarat it was 1139 per ha and 127 per ha for the dense and sparse mangroves respectively. Considering all the three density classes together, the tree density of Kachchh, Gulf of Kachchh, Saurashtra and South Gujarat comes out to be 8236 per, 3252 per ha, 2746 per ha and 723 per ha respectively. Considering four mangrove regions together, the tree density of mangroves in Gujarat is 8132 per ha, 5114 per ha and 2535 per ha for the dense, moderate and sparse mangroves respectively giving an average tree density of mangroves in the state to the tune of 5893 per ha.

Of the total mangrove trees, 97.62 % belongs to a single species i.e. *A. marina*. The rest 2.38% is collectively represented by other mangrove species (Table 5). Considering the height classes, 79.3% trees belong to the height class 1.5 to 3 m (Table 5). Further, 17.9%, 2.6% and 0.2% trees belong to the height class 3 to 4 m, 4 to 5 m and more than 5 m respectively.

The carbon sequestration potentials of different height classes of the nine mangrove species are given at Table 6. In case of recruits and plants up to 3 m height, it was maximum for *R. mucronata* and minimum for *A. officinalis* (Table 6). The recruits of *L. racemosa* were not reported from the state. For the height class 3 to 4 m, carbon sequestration potential was maximum for *R. mucronata* while minimum for *L. racemosa* (Table 6).

The mangrove plants (including recruits) have sequestered 2.24 million ton of carbon. Considering the height classes, 0.36 million ton (16.40%), 0.87 million ton (38.83%), 0.71 million ton (31.76%), 0.23 million ton (10.26%) and 0.06 million ton (2.7%) carbon has been sequestered by up to 1.5 m, 1.5 to 3 m, 3 to 4 m, 4 to 5 m and more than 5 m (Table 7). Further, 1.7 million ton (78%), 0.21 million ton (10%), 0.02 million ton (1%) and 0.25 million ton (11%) carbon has been sequestered by mangroves of Kachchh, Gulf of Kachchh, Saurashtra and South Gujarat respectively (Table 8).

In terms of carbon sequestration by mangrove plants (including recruits) per unit area, it is 31 ton per ha, 12 ton per ha, 7 ton per ha and 18 ton per ha for Kachchh, Gulf of Kachchh, Saurashtra and South Gujarat respectively. Considering the density classes, it comes out to be 36 ton per ha, 19 ton per ha and 10 ton per ha for dense moderate and sparse mangroves. In the case of mangrove soil, carbon sequestration was found more in the lower layer (16 to 30 cm depth) as compared to the upper layer (up to 15 cm depth). Considering the density classes of four mangrove regions, the carbon sequestered by mangrove soils is 4.01 million ton, 0.73 million ton and 1.14 million ton (Table 8) by the dense, moderate and sparse mangroves. Hence, a total of 5.87 million ton carbon has been sequestered by mangrove soils (up to 30 cm depth) of Gujarat (Table 8). Considering the four mangrove regions, 2.05 million ton (35%), 1.23 million ton (21%), 0.26 million ton (4%) and 2.33 million ton (40%) carbon has been sequestered by mangroves of Kachchh, Gulf of Kachchh, Saurashtra and South Gujarat respectively (Table 8).

The mangrove soil and plants (including recruits) together has sequestered 8.116 million ton carbon in Gujarat (Table 8). Out of this 3.793 million ton, 1.443 million ton, 0.288 million ton and 2.592 million ton has been sequestered by

Kachchh, Gulf of Kachchh, Saurashtra and South Gujarat respectively (Table 8). Considering the three density classes of the four mangrove regions, 5.629 million ton, 1.092 million ton and 1.395 million ton carbon has been sequestered by dense, moderate and sparse mangroves of the state (Table 8). Further, the carbon sequestration per unit area comes out to be 67.73 ton per ha, 82.90 ton per ha, 83.42 ton per ha and 180.24 ton per ha for Kachchh, Gulf of Kachchh, Saurashtra and South Gujarat respectively. Considering the four mangrove regions together, it comes out to be 88.95 ton per ha. In terms of three density classes, the carbon sequestration per unit area is 123.30 ton per ha, 55.35 ton per ha and 53.95 ton per ha for the dense, moderate and sparse mangroves respectively.

In terms of carbon dioxide, 29.76 million ton has been sequestered by mangrove plants (including recruits) and soil (up to 30 cm depth) together. In terms of carbon dioxide sequestered per unit area, it has been found to be 248.3 t ha<sup>-1</sup>, 304.0 t ha<sup>-1</sup>, 305.9 t ha<sup>-1</sup> and 660.9 t ha<sup>-1</sup> by Kachchh, Gulf of Kachchh, Saurashtra and South Gujarat mangroves respectively. Considering the density classes of all the four mangrove zones, the carbon dioxide sequestered per unit areas has been estimated as 452.1 t ha<sup>-1</sup>, 203.0 t ha<sup>-1</sup> and 197.8 t ha<sup>-1</sup> by the dense, moderate and sparse mangrove respectively (Table 9).

## DISCUSSIONS AND CONCLUSIONS

The recruit density of mangroves of Philippines has been estimated to range from 2,533±1,139 to 1,96,000±8,461 per ha (Antonio, et. al., 1999). The recruit density of the dwarf and fringe mangrove forests of Biscayne National Park, Florida (U.S.A.) has been estimated as 6,08,200±1,22,100 per ha and 1,07,200±42,900 per ha, respectively (Michael, et. al., 2001). (Nayar, 2008) has found that for Kerala mangroves the recruit density is highest for *Aegiceras corniculatum* (1,35,229 per ha) which is followed by *Bruguiera cylindrical* (42,823 per ha), *Acanthus ilicifolius* (14,414 per ha), *Avicennia officinalis* (12,629 per ha), *Excoecaria agallocha* (7,143 per ha), *Avicennia marina* (5,714 per ha), *Rhizophora mucronata* (1,886 per ha), *Kandelia candel* (943 per ha) and *R. apiculata* (471 per ha). The recruit density of mangrove estimated by the present work is within the range of available information.

The tree density of mangroves of Philippines has been estimated as 3,300±656 trees per ha (Antonio, et. al., 1999). For the dwarf and fringe mangrove forests of Biscayne National Park, Florida (U.S.A.), it has been estimated as 8,800±8,800 trees per ha and 57,800±9,000 trees per ha respectively (Michael, et. al., 2001). The tree density of Bahamas, Jamanica, Puerto Rico, Belize Venezuela and Bon Accord Lagoon mangrove forests of Tobago (of *R. mangle*) has been estimated as 1140 per ha, 4950 per ha, 2370 per ha, 4520 per ha, 2630 per ha and 4620 per ha respectively. Further, the tree density of mangroves of Bertioga and Guaratiba, Southeast Brazil estimated as 4100 trees per ha and 2560 trees per ha (average tree height 7 m) where the average annual rainfall was 1032mm and 2240mm respectively (Soares & Yara, 2005). The average annual rain fall ranges from 2000 mm (South Gujarat) to less than 400 mm (Kachchh) in the state. The present work has estimated the tree density for Kachchh, Gulf of Kachchh, Saurashtra and South Gujarat as 8,236 per ha, 3252 per ha, 2,746 per ha and 723 per ha respectively. However, considering all the four mangrove regions together, it is 5893 per ha. Further, in terms of three density classes, the present work has found the tree density as 8132 per ha, 5114 per ha and 2535 per ha for dense, moderate and sparse mangroves respectively. The tree density of *A. marina* for the mangroves of French Guiana and Northern Australia has been reported to be 1,06,000 per ha (0.6m height), 8,831 per ha (2.7 m height) and 7,500 per ha (3.4 m height) (Christophe, et. al., 2003).

The biomass carbon content values range from 25 t CO<sub>2</sub> ha<sup>-1</sup> to 2,254 t CO<sub>2</sub> ha<sup>-1</sup> and most estimates fall between 300 and 1,000 t CO<sub>2</sub> ha<sup>-1</sup> (Samantha, et. al., 2011). Considering the vegetation carbon pool, the CO<sub>2</sub> sequestered per unit area in Kachchh, Gulf of Kachchh, Saurashtra and South Gujarat has been found to be 308 t ha<sup>-1</sup>, 95.3 t ha<sup>-1</sup>, 55.0 tons ha<sup>-1</sup>

and 117.3 t ha<sup>-1</sup> respectively. All the mangrove regions put together, the CO<sub>2</sub> sequestered per unit area has been found to be 143.9 t ha<sup>-1</sup> for the mangroves of Gujarat. In terms of the three density classes, the carbon dioxide sequestered by mangrove plants (including recruits) is 95.3 t ha<sup>-1</sup>, 39.1 t ha<sup>-1</sup> and 19.3 t ha<sup>-1</sup> for the dense, moderate and sparse mangroves of Gujarat.

Tropical wetland forests such as mangroves are important reservoirs of organic C as the soil depth is more (Page, et. al., 2001; Page, 2002; Murdiyarso, 2010). High carbon organic matter (as high as 75% carbon concentration) can be found as deep as 8m belowground (Chmura, et. al., 2003). Carbon accumulation rates in mangroves are 10 times the rate for temperate forests and up to 50 times the rate for tropical forests (Laffoley, et. al., 2009). In other words, 1 km<sup>2</sup> of mangrove area results in the equivalent long term sequestration in 50 km<sup>2</sup> of tropical forests (Yee., 2010).

Peatlands received significant attention since 1997 (Duke, 2007), when peat fires associated with land clearing in Indonesia increased atmospheric CO<sub>2</sub> enrichment by 13 to 40% over global annual fossil fuel emissions (Page, 2002). It has resulted in incorporation of peatlands in the international climate change mitigation strategies (Murdiyarso, et. al., 2010). However, mangroves were not given due importance in such strategies although it occur in 118 countries, adding 30 to 35% to the global area of tropical wetland forest over peat swamps alone (FAO, 2007; Page, et. al., 2001).

Though mangroves are well known for high C assimilation and flux rates (Twilley, et. al., 1992; Chmura, et. al., 2003; Alongi, et. al., 2004; Komiyama, et. al., 2008; Kristensen, et. al., 2008), data are surprisingly lacking on whole-ecosystem carbon storage - the amount which stands to be released with land-use conversion. Limited components of carbon storage have been reported, most notably tree biomass (Twilley, et. al., 1992; Kristensen, et. al., 2008), but evidence of deep organic-rich soils (Golley, 1962; Matsui, 1998; Fujimoto, 1999) suggests these estimates miss the vast majority of total ecosystem carbon. These facts have been re-established by the present work which has found that of the total carbon sequestered by mangroves of Gujarat, 72% is by soil and 28% is by mangrove vegetation.

In terrestrial ecosystems the soil organic carbon shows a decreasing trend with the depth of soil (Yang, et. al., 2010). However, this is not the case with mangroves. Mangrove soils consist of a variably thick, tidally submerged suboxic layer (variously called 'peat' or 'muck') supporting anaerobic decomposition pathways and having moderate to high C concentration (Kristensen, et. al., 2008). The present work has also found that the soil organic carbon is more in the lower layer (16 to 30 cm) than the upper one (up to 15 cm depth).

Daniel, (2011) has examined 25 mangrove sites (n =10 estuarine, n = 15 oceanic) across the Indo-Pacific (8° S-22° N, 90° -163° E) and found high carbon concentration (% dry mass) throughout the top 1 m of the soil profile, with a decline below 1m. They have reported that mangroves are among the most carbon dense forests in the tropics (mean 1,023 t C ha<sup>-1</sup>), and exceptionally high compared to mean carbon storage of the world's major forest domains (Boreal- ~350 t C ha<sup>-1</sup>, Temperate ~350 t C ha<sup>-1</sup>, upland~300 t C ha<sup>-1</sup>). Estuarine sites contained a mean of 1,074 t C ha<sup>-1</sup>; oceanic sites contained 990±96 t C ha<sup>-1</sup>. Above-ground carbon pools were sizeable (mean 159 t C ha<sup>-1</sup>, maximum 435 t C ha<sup>-1</sup>), but below-ground storage in soils dominated, accounting for 71-98% and 49-90% of total storage in estuarine and oceanic sites, respectively (Daniel, 2011). Further, Ong (2002) has reported that the sediments in mangrove forests held 700 tons of carbon per meter depth per hectare.

The SOC of mangrove soil of (*R. apiculata* crop) at Tambol Yisan, Samut Songkram Province has been estimated as 50-70 t ha<sup>-1</sup> for a 9-10 yr old plantation while for younger plantation it was reported as 50 t ha<sup>-1</sup> (Kridiborworn, unpublished data). The present work has estimated 87.83 t ha<sup>-1</sup>, 36.99 t ha<sup>-1</sup> and 44.08 t ha<sup>-1</sup> carbon sequestration for dense, moderate and sparse mangroves of Gujarat.

Considering the four mangrove regions, the SOC was found  $36.60 \text{ t ha}^{-1}$ ,  $70.64 \text{ t ha}^{-1}$ ,  $75.31 \text{ t ha}^{-1}$  and  $162.02 \text{ t ha}^{-1}$  for Kachchh, Gulf of Kachchh, Saurashtra and South Gujarat. The Indian council of Forestry Research and Education has estimated the SOC for 14 types of Forests of India among them the soils of littoral swamp forests of have been reported to sequester the highest carbon i.e.  $155.22 \text{ t ha}^{-1}$  (Kishwan., 2009).

The SOC of mangrove forests of Khanom district of Thailand has been found to increase from  $110 \text{ t ha}^{-1}$  to  $160 \text{ t ha}^{-1}$  after hydraulic restoration (by opening shrimp pond bank) which was very low as compared to the natural mangrove forests i.e.  $370 \text{ t ha}^{-1}$  to  $553 \text{ t ha}^{-1}$  (Matsui, et. al., 2010). Higher temperatures and wet environments result in increased decomposition rates in wetlands soils (Chmura, et. al., 2003). Efficiency of carbon sequestration in sediments improves with the age of the mangrove forests, from 16% for a 5-year old stand to 27% for an 85 year old stand (Alongi, 2004; Lafolley & Grimsditch, 2009). However, mangrove SOC is susceptible to decomposition due to the abundance of aliphatic-rich humic acid which easily change to carboxylic humic substances by humification (Matsui, 2007; Orlov, 1995; Yonebayashi, 1994). This intrinsic character of mangrove SOC may lead to accelerated carbon decomposition in the soil which is exposed to air (Matsui, et. al., 2010). Therefore, inappropriate land use in mangrove area risks including a significant carbon loss transforming mangrove areas to source of  $\text{CO}_2$  emission. In the mangrove areas which receive proper and adequate inundation ( $>700$  hrs per year), the SOC is protected from severe decomposition (Matsui, et. al., 2010). However, in the mangrove areas receiving  $<200$  hr per year inundation, the SOC is very low (Matsui, et. al., 2010).

The carbon sequestration by Matang mangrove forests of Malaysia has been estimated as  $1.5 \text{ t h}^{-1} \text{ Yr}^{-1}$ . The above carbon pool in 25 mangrove systems across the Indo-Pacific has been estimated as  $159 \text{ t ha}^{-1}$ . The carbon sequestered by undisturbed mangroves of Términos Lagoon, Campeche has been estimated as  $396.1 \text{ t ha}^{-1}$  (Cerón-Bretón, et. al., 2010). The carbon sequestered by the mangrove forests of Everglades National Park, South Florida, USA has been estimated as  $5.6 \times 10^9 \text{ kg}$  or 5.6 million tons (Marc, et. al., 2006). The present work has reported that carbon sequestered in the mangrove forests (plants and soil) of Gujarat is 8.116 million ton.

Kishwan. J. (2009) has estimated carbon sequestration per unit area by (littoral and) swam forests of India as  $106.9 \text{ t ha}^{-1}$ . Present work has estimated the carbon sequestration per unit area by Kachchh, Gulf of Kachchh, Saurashtra and South Gujarat as  $67.73 \text{ t ha}^{-1}$ ,  $82.90 \text{ t ha}^{-1}$ ,  $83.42 \text{ t ha}^{-1}$  and  $180.24 \text{ t ha}^{-1}$  respectively. In terms of dense, moderate and sparse mangrove forests of the state, it is  $123.30 \text{ t ha}^{-1}$ ,  $55.35 \text{ t ha}^{-1}$  and  $53.95 \text{ t ha}^{-1}$  respectively.

## ACKNOWLEDGMENTS

We acknowledge GEER Foundation, Gandhinagar for providing necessary support for implementing the project. We also acknowledge the Ministry of Forest and Environment, Gov. of Gujarat for providing financial assistance. We are also thankful to the officials and staff of Gujarat Forest department for providing the local help and the permissions for field works. The thanks are also due to Dr. Harshad Salvi (Research associate, GEER Foundation), Mr. Mukesh Mali (Senior Research Fellow, GEER Foundation), Mr. Bipin Khokhariya (Senior Research Fellow, GEER Foundation), Mr. Bhargav Brahmbhatt (Junior Research Fellow, GEER Foundation), Ms. Reshma Bobeda (Senior Research Fellow, GEER Foundation), Ms. Jagruti Jaggiwala (Junior Research Fellow, GEER Foundation), for helping field works.

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#### Tables for the Paper Carbon Sequestration by Mangroves of Gujarat

**Table 1: Occurrence of Mangrove Species in Gujarat**

| Sr. No. | Species                                     | Relative Occurrence in the State                   | Reported Location                       | Selected for Uprooting |
|---------|---------------------------------------------|----------------------------------------------------|-----------------------------------------|------------------------|
| 1       | <i>Avicennia alba</i> Blume                 | Rare                                               | South Gujarat                           | No                     |
| 2       | <i>Avicennia marina</i> (Forsk.) Vierh.     | Dominates the mangrove of Gujarat                  | All the four mangrove regions           | Yes                    |
| 3       | <i>Avicennia officinalis</i> L.             | Rare                                               | South Gujarat                           | Yes                    |
| 4       | <i>Rhizophora apiculata</i> Bl.             | Extremely rare                                     | South Gujarat                           | No                     |
| 5       | <i>Rhizophora mucronata</i> Lamk            | Common in the Gulf of Kachchh, rare in other areas | Gulf of Kachchh and South Gujarat       | Yes                    |
| 6       | <i>Acanthus ilicifolius</i> L.              | Rare                                               | South Gujarat                           | Yes                    |
| 7       | <i>Ceriops decandra</i> (Griff.) Ding Hou   | Extremely rare (Only a few trees)                  | South Gujarat.                          | No                     |
| 8       | <i>Ceriops tagal</i> (Perr.) C. B. Robinson | Common in the Gulf of Kachchh, rare in other areas | Kachchh, Gulf of Kachchh, South Gujarat | Yes                    |
| 9       | <i>Bruguiera cylindrica</i> (L.) Bl.        | Rare (More than 100 reproductively matured trees)  | South Gujarat                           | Yes                    |
| 10      | <i>Bruguiera gymnorrhiza</i> (L.) Lamk      | Rare (less than 50 trees)                          | South Gujarat                           | No                     |
| 11      | <i>Sonneratia apetala</i> Buch. – Ham.      | Common in South Gujarat                            | South Gujarat                           | Yes                    |
| 12      | <i>Excoecaria agallocha</i> L.              | Rare (less than 100 trees)                         | South Gujarat                           | No                     |
| 13      | <i>Aegiceras corniculatum</i> (L.) Blanco   | Rare                                               | Gulf of Kachchh, South Gujarat          | Yes                    |
| 14      | <i>Kandelia candel</i> (L.) Druce           | Rare (Only one tree)                               | South Gujarat                           | No                     |
| 15      | <i>Lumnitzera racemosa</i> Willd            | Rare (About 200 trees)                             | South Gujarat                           | Yes                    |

**Table 2: Distribution of Mangrove Cover (ha) in the Four Regions of Gujarat**

| Regions         | Dense          | Moderate     | Sparse       | Total          | % Distribution |
|-----------------|----------------|--------------|--------------|----------------|----------------|
| Kachchh         | 25798          | 14370        | 15836        | 56004          | 61 %           |
| Gulf of Kachchh | 9275           | 4541         | 3595         | 17411          | 19 %           |
| Saurashtra      | 2106           | 823          | 523          | 3452           | 4 %            |
| South Gujarat   | 8474.5         | 0            | 5906.2       | 14380.7        | 16 %           |
| <b>Total</b>    | <b>45653.5</b> | <b>19734</b> | <b>25860</b> | <b>91247.7</b> | <b>100 %</b>   |

**Table 3: Details of No. of Plots Laid for Mangrove Population Estimation and No. of Soil Samples Collected for SOC Estimation**

| Mangrove Regions | No. of Plots Laid for Mangrove Population Estimation | No. of Soil Samples Collected |
|------------------|------------------------------------------------------|-------------------------------|
| Kachchh          | 120                                                  | 76                            |
| Gulf of Kachchh  | 101                                                  | 0                             |
| Saurashtra       | 39                                                   | 24                            |
| South Gujarat    | 56                                                   | 36                            |
| <b>Total</b>     | <b>316</b>                                           | <b>136</b>                    |

**Table 4: Estimated No. of Mangrove Recruits in Gujarat**

| Mangrove Regions | Dense             | Moderate         | Sparse           | Total             | % Distribution |
|------------------|-------------------|------------------|------------------|-------------------|----------------|
| Kachchh          | 430950430         | 169393560        | 65548371         | 665892362         | 49%            |
| Gulf of Kachchh  | 53122069          | 11453411         | 3287566          | 67863046          | 5%             |
| Saurashtra       | 22229708          | 6421645          | 1120528          | 29771880          | 2%             |
| South Gujarat    | 502789374         | NA               | 98150275         | 600939648         | 44%            |
| <b>Total</b>     | <b>1009091581</b> | <b>187268616</b> | <b>168106739</b> | <b>1364466935</b> | <b>100%</b>    |

**Table 5: Species Wise Details about No. of Mangrove Trees in Four Mangrove Regions**

| Species                | Mangrove Regions | 151-300 cm       | 301-400 cm      | 401-500 cm      | >500 cm       | Total            | % of sp.   | % of all sp. |
|------------------------|------------------|------------------|-----------------|-----------------|---------------|------------------|------------|--------------|
| <i>A. marina</i>       | Kachchh          | 356092990        | 92438016        | 12368662        | 0             | 460899668        | 88         | 97.62        |
|                        | GOK              | 43403787         | 1443696         | 157872          | 175633        | 45180988         | 9          |              |
|                        | Saurashtra       | 9477983          | 0               | 0               | 0             | 9477983          | 2          |              |
|                        | South Gujarat    | 5113035          | 2178295         | 1267216         | 794046        | 9352592          | 1          |              |
|                        | <b>Total</b>     | <b>414087795</b> | <b>96060007</b> | <b>13793750</b> | <b>969679</b> | <b>524911231</b> | 100        |              |
| <i>A. officinalis</i>  | Kachchh          | 0                | 0               | 0               | 0             | 0                | 0          | 0.03         |
|                        | GOK              | 0                | 0               | 0               | 0             | 0                | 0          |              |
|                        | Saurashtra       | 0                | 0               | 0               | 0             | 0                | 0          |              |
|                        | South Gujarat    | 72581            | 32070           | 18567           | 11815         | 135033           | 100        |              |
|                        | <b>Total</b>     | <b>72581</b>     | <b>32070</b>    | <b>18567</b>    | <b>11815</b>  | <b>135033</b>    | <b>100</b> |              |
| <i>A. ilicifolius</i>  | Kachchh          | 0                | 0               | 0               | 0             | 0                | 0          | 0.04         |
|                        | GOK              | 0                | 0               | 0               | 0             | 0                | 0          |              |
|                        | Saurashtra       | 0                | 0               | 0               | 0             | 0                | 0          |              |
|                        | South Gujarat    | 226540           | 0               | 0               | 0             | 226540           | 100        |              |
|                        | <b>Total</b>     | <b>226540</b>    | <b>0</b>        | <b>0</b>        | <b>0</b>      | <b>226540</b>    | 100        |              |
| <i>A. corniculatum</i> | Kachchh          | 0                | 0               | 0               | 0             | 0                | 0          | 0.22         |
|                        | GOK              | 1172671          | 0               | 0               | 0             | 1172671          | 99         |              |
|                        | Saurashtra       | 0                | 0               | 0               | 0             | 0                | 0          |              |
|                        | South Gujarat    | 17529            | 0               | 0               | 0             | 17529            | 1          |              |
|                        | <b>Total</b>     | <b>1190200</b>   | <b>0</b>        | <b>0</b>        | <b>0</b>      | <b>1190200</b>   | <b>100</b> |              |
| <i>B. cylindrica</i>   | Kachchh          | 0                | 0               | 0               | 0             | 0                | 0          | 0.00         |
|                        | GOK              | 0                | 0               | 0               | 0             | 0                | 0          |              |
|                        | Saurashtra       | 0                | 0               | 0               | 0             | 0                | 0          |              |
|                        | South Gujarat    | 477              | 0               | 0               | 0             | 477              | 100        |              |
|                        | <b>Total</b>     | <b>477</b>       | <b>0</b>        | <b>0</b>        | <b>0</b>      | <b>477</b>       | <b>100</b> |              |

Table 5: Contd.,

| Table 5: Contd.,                             |               |           |          |          |         |           |     |      |
|----------------------------------------------|---------------|-----------|----------|----------|---------|-----------|-----|------|
| C. tagal                                     | Kachchh       | 329389    | 0        | 0        | 0       | 329389    | 4   | 1.41 |
|                                              | GOK           | 7201579   | 45388    | 0        | 0       | 7246967   | 96  |      |
|                                              | Saurashtra    | 0         | 0        | 0        | 0       | 0         | 0   |      |
|                                              | South Gujarat | 170       | 0        | 0        | 0       | 170       | 0   |      |
|                                              | Total         | 7531138   | 45388    | 0        | 0       | 7576526   | 100 |      |
| R. mucronata                                 | Kachchh       | 0         | 0        | 0        | 0       | 0         | 0   | 0.56 |
|                                              | GOK           | 3000935   | 23681    | 0        | 0       | 3024616   | 100 |      |
|                                              | Saurashtra    | 0         | 0        | 0        | 0       | 0         | 0   |      |
|                                              | South Gujarat | 1         | 0        | 0        | 0       | 1         | 0   |      |
|                                              | Total         | 3000936   | 23681    | 0        | 0       | 3024617   | 100 |      |
| L. racemosa                                  | Kachchh       | 0         | 0        | 0        | 0       | 0         | 0   | 0.00 |
|                                              | GOK           | 0         | 0        | 0        | 0       | 0         | 0   |      |
|                                              | Saurashtra    | 0         | 0        | 0        | 0       | 0         | 0   |      |
|                                              | South Gujarat | 0         | 200      | 0        | 0       | 200       | 100 |      |
|                                              | Total         | 0         | 200      | 0        | 0       | 200       | 100 |      |
| S. apetala                                   | Kachchh       | 0         | 0        | 0        | 0       | 0         | 0   | 0.12 |
|                                              | GOK           | 0         | 0        | 0        | 0       | 0         | 0   |      |
|                                              | Saurashtra    | 0         | 0        | 0        | 0       | 0         | 0   |      |
|                                              | South Gujarat | 346720    | 154763   | 126225   | 41217   | 668925    | 100 |      |
|                                              | Total         | 346720    | 154763   | 126225   | 41217   | 668925    | 100 |      |
| Total Mangrove Trees                         |               | 426456387 | 96316109 | 13938542 | 1022711 | 537733749 | NA  |      |
| % Distribution Over Different Height Classes |               | 79.3%     | 17.9     | 2.6%     | 0.2     | 100.0     | NA  |      |

Table 6: Height Class Wise Details of Carbon Sequestration Potential (kg) of Nine Mangroves Species

| Species                | 1-150 cm | 151-300 cm | 301-400 cm | 401-500 cm | > 500 cm |
|------------------------|----------|------------|------------|------------|----------|
| <i>A. marina</i>       | 0.26     | 1.91       | 7.39       | 16.56      | 55.84    |
| <i>A. officinalis</i>  | 0.01     | 0.28       | 3.72       | 6.17       | 65.66    |
| <i>S. apetala</i>      | 0.18     | 0.90       | 8.09       | 14.61      | 146.25   |
| <i>A. ilicifolius</i>  | 0.61     | 0.88       | NA         | NA         | NA       |
| <i>A. corniculatum</i> | 0.23     | 1.98       | NA         | NA         | NA       |
| <i>B. cylindrica</i>   | 0.22     | 0.29       | NA         | NA         | NA       |
| <i>C. tagal</i>        | 1.00     | 4.27       | 9.97       | NA         | NA       |
| <i>R. mucronata</i>    | 1.60     | 14.69      | 46.32      | NA         | NA       |
| <i>L. racemosa</i>     | NA       | NA         | 0.96       | NA         | NA       |

Table 7: Carbon Sequestration (Million Ton) by Mangrove Plants of Different Height Classes of Four Mangrove Regions

| Mangrove Region | 1-150 cm      | 151-200 cm    | 301-400 cm    | 401-500 cm    | >500 cm       | Total         | % Distribution |
|-----------------|---------------|---------------|---------------|---------------|---------------|---------------|----------------|
| Kachchh         | 0.1753        | 0.6820        | 0.6824        | 0.2047        | 0.0000        | 1.7444        | <b>78</b>      |
| GOK             | 0.0297        | 0.1601        | 0.0122        | 0.0026        | 0.0098        | 0.2144        | <b>10</b>      |
| Saurashtra      | 0.0077        | 0.0181        | 0.0000        | 0.0000        | 0.0000        | 0.0258        | <b>1</b>       |
| South Gujarat   | 0.1551        | 0.0103        | 0.0175        | 0.0229        | 0.0512        | 0.2570        | <b>11</b>      |
| <b>Total</b>    | <b>0.3678</b> | <b>0.8705</b> | <b>0.7121</b> | <b>0.2302</b> | <b>0.0610</b> | <b>2.2416</b> | <b>100</b>     |

Table 8: Carbon Sequestered in Mangrove Forests (million ton)

| Regions         | Particulars  | Dense        | Moderate     | Sparse       | All Density Classes |
|-----------------|--------------|--------------|--------------|--------------|---------------------|
| Kachchh         | Soils        | 1.179        | 0.430        | 0.440        | 2.049               |
|                 | Plants       | 1.180        | 0.344        | 0.220        | 1.744               |
|                 | <b>Total</b> | <b>2.360</b> | <b>0.774</b> | <b>0.660</b> | <b>3.793</b>        |
| Gulf of Kachchh | Soils        | 0.912        | 0.250        | 0.066        | 1.229               |
|                 | Plants       | 0.190        | 0.021        | 0.005        | 0.215               |
|                 | <b>Total</b> | <b>1.102</b> | <b>0.271</b> | <b>0.071</b> | <b>1.443</b>        |

| Table 8: Contd., |              |              |              |              |              |
|------------------|--------------|--------------|--------------|--------------|--------------|
| Saurashtra       | Soils        | 0.207        | 0.045        | 0.010        | 0.262        |
|                  | Plants       | 0.023        | 0.003        | 0.001        | 0.026        |
|                  | <b>Total</b> | <b>0.230</b> | <b>0.048</b> | <b>0.010</b> | <b>0.288</b> |
| South Gujarat    | Soils        | 1.709        | NA           | 0.626        | 2.335        |
|                  | Plants       | 0.228        | NA           | 0.029        | 0.257        |
|                  | <b>Total</b> | <b>1.937</b> | <b>0.000</b> | <b>0.655</b> | <b>2.592</b> |
| All regions      | Soils        | <b>4.008</b> | <b>0.725</b> | <b>1.141</b> | <b>5.874</b> |
|                  | Plants       | <b>1.621</b> | <b>0.367</b> | <b>0.254</b> | <b>2.242</b> |
|                  | <b>Total</b> | <b>5.629</b> | <b>1.092</b> | <b>1.395</b> | <b>8.116</b> |

Table 9: CO<sub>2</sub> Sequestered per Unit Area (Ton per ha)

| CO <sub>2</sub> Sequestered per Unit Area (Ton per ha) |       |          |        |                                |
|--------------------------------------------------------|-------|----------|--------|--------------------------------|
| Regions                                                | Dense | Moderate | Sparse | Average of All Density Classes |
| Kachchh                                                | 335.4 | 197.4    | 152.7  | 248.3                          |
| Gulf of Kachchh                                        | 435.7 | 218.5    | 72.1   | 304.0                          |
| Saurashtra                                             | 400.3 | 213.4    | 71.1   | 305.9                          |
| South Gujarat                                          | 838.2 | NA       | 406.5  | 660.9                          |
| Avg. of all zones                                      | 452.1 | 203.0    | 197.8  | 326.1                          |